



Mixed-Integer Linear Programming

Prof. Karl Ezra Pilario

EgyE 332

1s AY 2026-2027

Outline

1. Motivational Example
2. Branch-and-Bound
3. Examples in Energy Engineering
 - a. Simple Unit Commitment
 - b. Unit Commitment with Renewables and Storage
 - c. Waste-to-Energy Facility Location-Allocation

Mixed-Integer Linear Programming

Energy Investment Portfolio Selection

https://github.com/kspilario/EgyE332_opti/blob/main/MILP%20Problems/energy_investment_knapsack.ipynb

An energy company has a limited capital budget and must decide which candidate energy projects to invest in.

Each project has:

- a fixed investment cost, and
- an expected annual energy production.

A project is either **selected completely** or **not selected at all**. Partial investment is not allowed.

The objective is to select the combination of projects that **maximizes total annual energy production** without exceeding the available investment budget.

Project	Technology	Investment Cost (₱ million)	Annual Energy Production (GWh/year)
1	Solar Farm	40	55
2	Wind Farm	50	75
3	Small Hydro	60	85
4	Biomass Plant	35	45
5	Rooftop Solar Program	25	30

Budget: PhP 100 million



$$\begin{aligned} \max \quad & 55x_1 + 75x_2 + 85x_3 + 45x_4 + 30x_5 \\ \text{s. t.} \quad & 40x_1 + 50x_2 + 60x_3 + 35x_4 + 25x_5 \leq 100 \\ & x_i \in \{0, 1\} \end{aligned}$$

This is an 0-1 Knapsack Problem.

Mathematical Programs

In general, the method for solving mathematical models is highly dependent on the **type of model**:

TYPE	Objective Function	Constraints	Decision Variables
Linear Program (LP)	Linear	Linear	Continuous
Nonlinear Program (NLP)	At least the objective function or 1 constraint is nonlinear		Continuous
Integer Program (IP)	Linear or nonlinear	Linear or nonlinear	All Integer
Mixed-integer Linear Program (MILP)	Linear	Linear	Some Integer; Others Continuous
Mixed-integer Nonlinear Program (MINLP)	At least the objective function or 1 constraint is nonlinear		Some Integer; Others Continuous

We'll discuss
**Branch-and-Bound
Method.**

Source: Chap.14; El-Halwagi, M., 2012. *Sustainable Design Through Process Integration*. USA: Elsevier Inc.

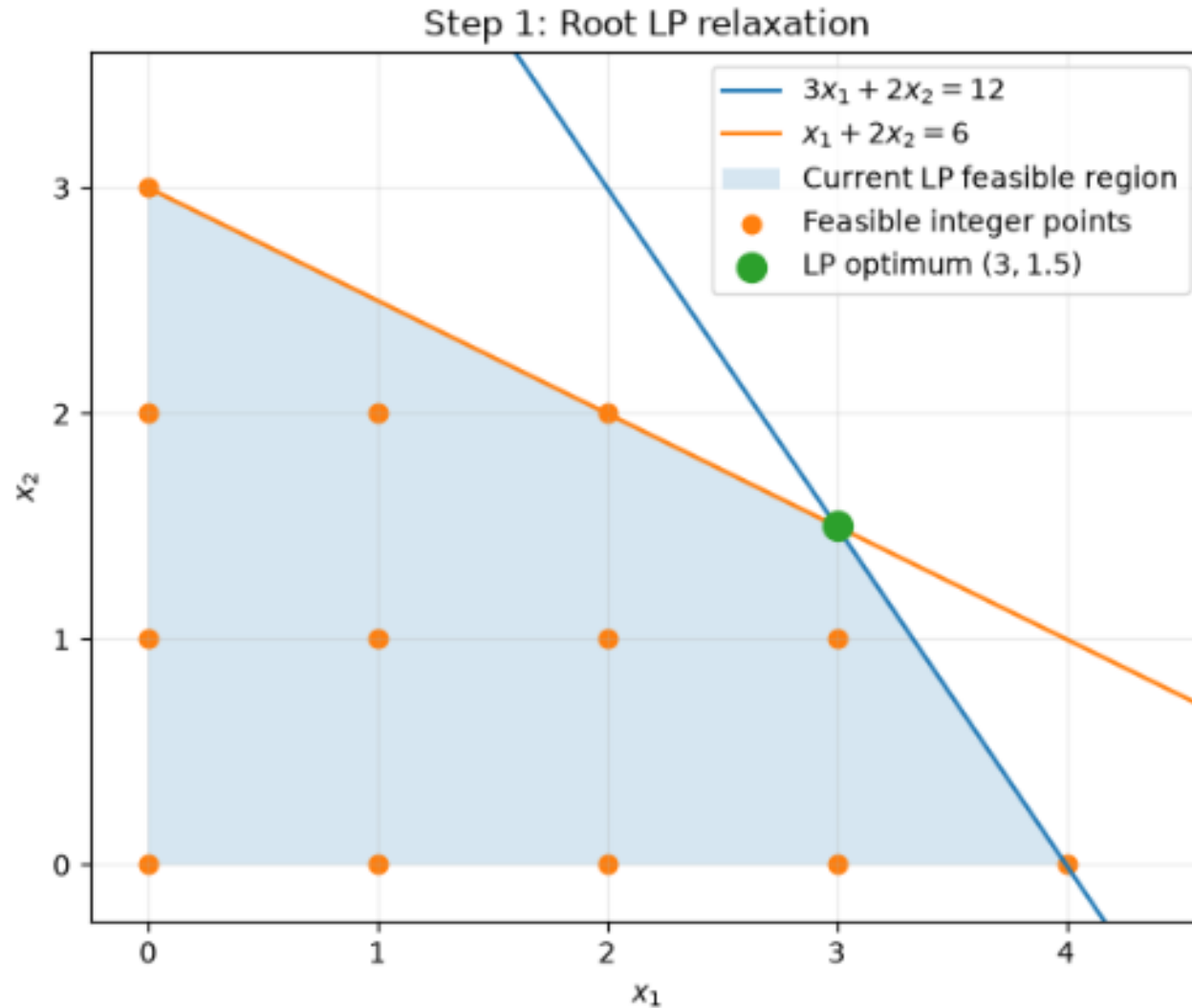
*The word "program" is due to George Dantzig (1974).

Recall: Linear Programming

Consider an **integer program**:

$$\begin{array}{ll}\max & 5x_1 + 4x_2 \\ \text{s.t.} & 3x_1 + 2x_2 \leq 12 \\ & x_1 + 2x_2 \leq 6 \\ & x_1, x_2 \in \mathbb{Z}^+\end{array}$$

But for now, what if we let x_1, x_2 be continuous and solve the **LP** first?



Branch-and-Bound Method

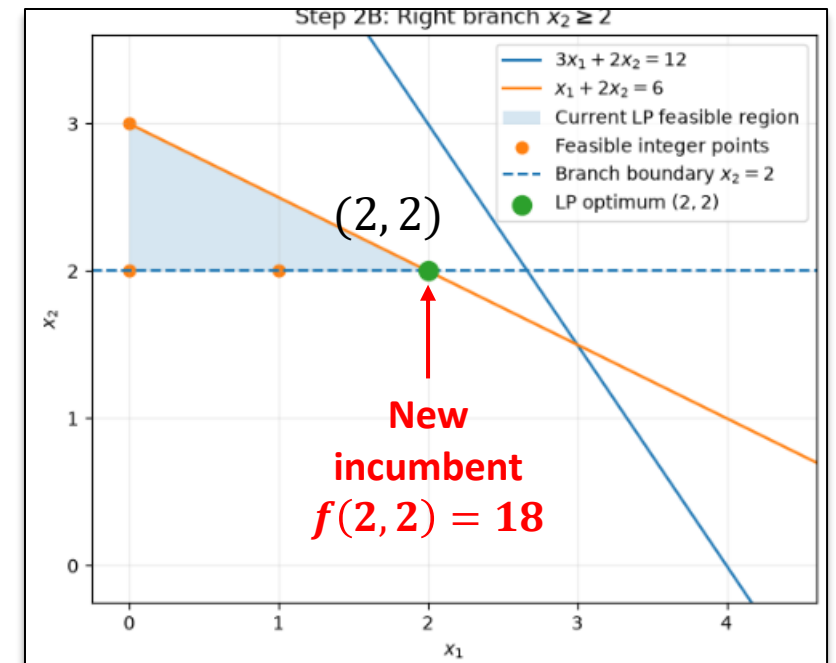
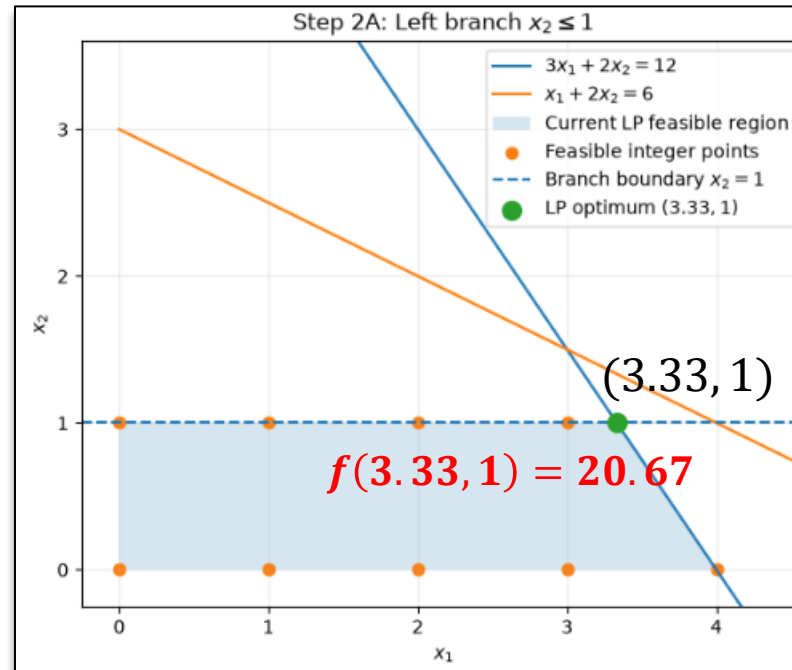
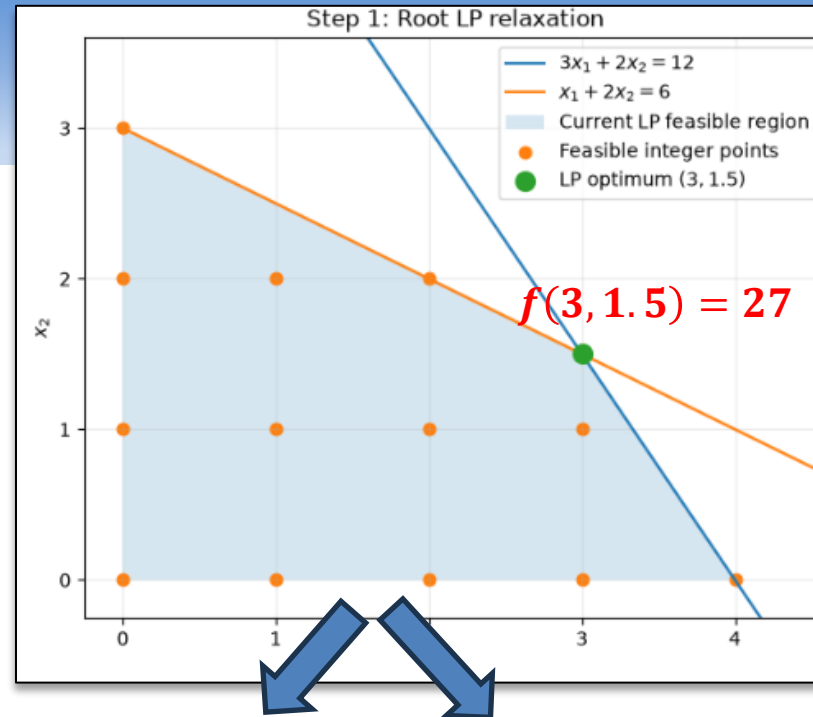
Consider an **integer program**:

$$\begin{aligned} \max \quad & 5x_1 + 4x_2 \\ \text{s.t.} \quad & 3x_1 + 2x_2 \leq 12 \\ & x_1 + 2x_2 \leq 6 \\ & x_1, x_2 \in \mathbb{Z}^+ \end{aligned}$$

In branching, we split the non-integer solution into two, then solve the LP in each.

$$x_2 \leq 1$$

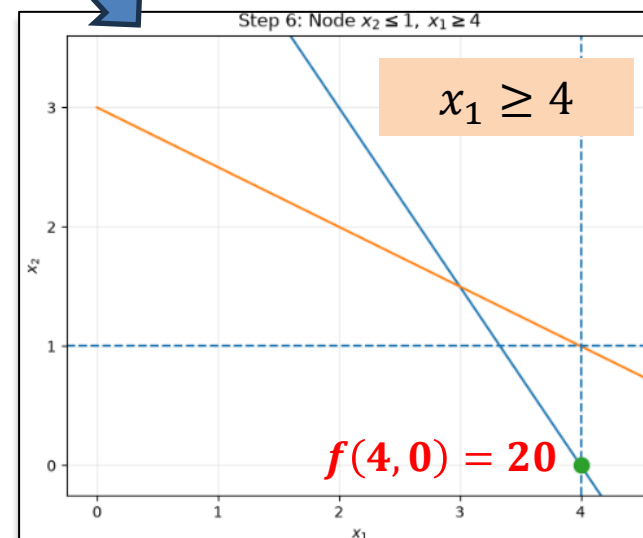
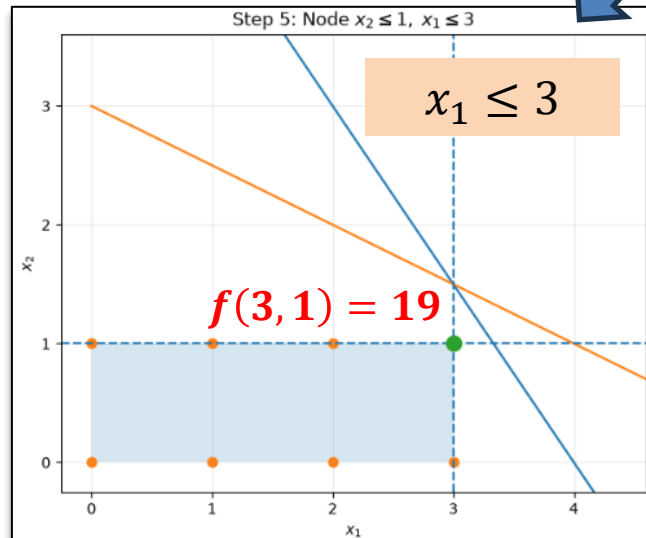
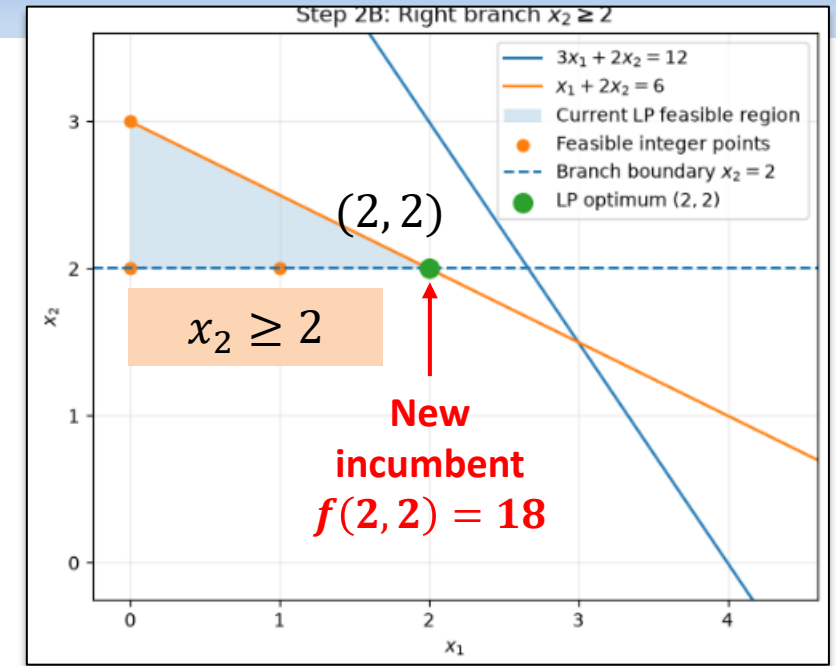
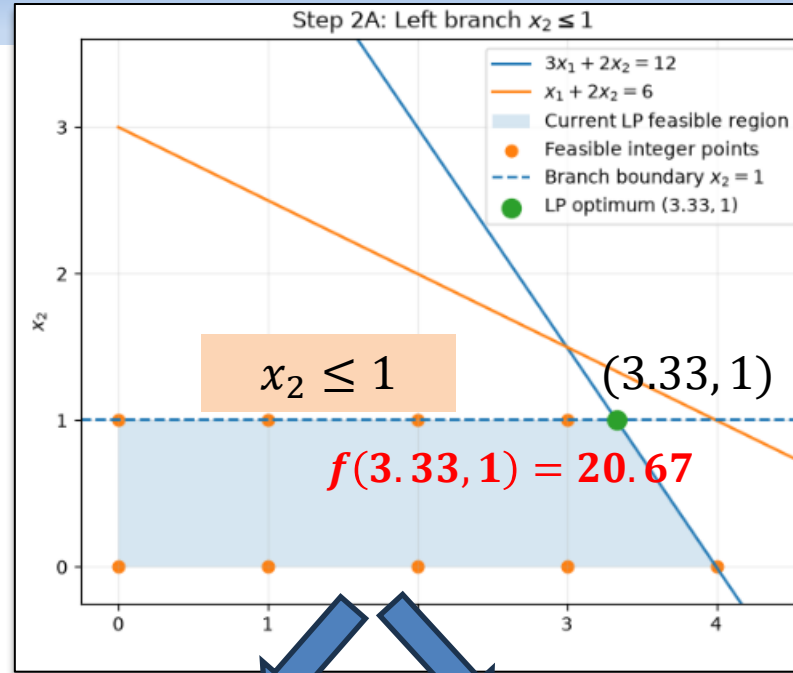
$$x_2 \geq 2$$



Branch-and-Bound Method

Consider an **integer program**:

$$\begin{aligned} \max \quad & 5x_1 + 4x_2 \\ \text{s.t.} \quad & 3x_1 + 2x_2 \leq 12 \\ & x_1 + 2x_2 \leq 6 \\ & x_1, x_2 \in \mathbb{Z}^+ \end{aligned}$$



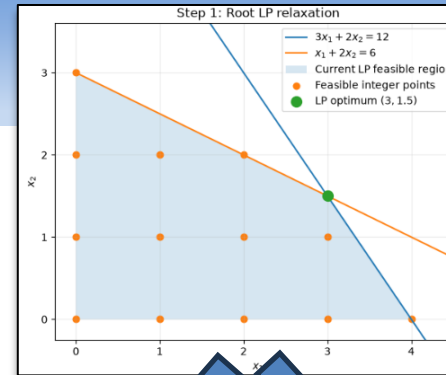
We branch on the left and right of $(3.33, 1)$ since it still has a fractional part.

Branch-and-Bound Method

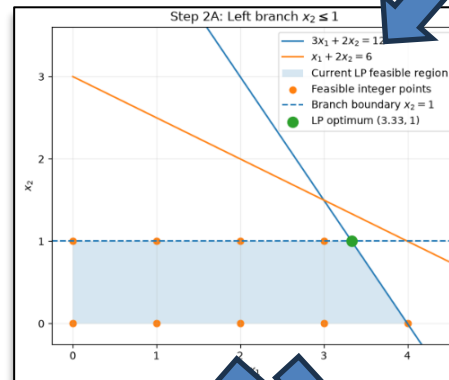
Consider an **integer program**:

$$\begin{aligned} \max \quad & 5x_1 + 4x_2 \\ \text{s.t.} \quad & 3x_1 + 2x_2 \leq 12 \\ & x_1 + 2x_2 \leq 6 \\ & x_1, x_2 \in \mathbb{Z}^+ \end{aligned}$$

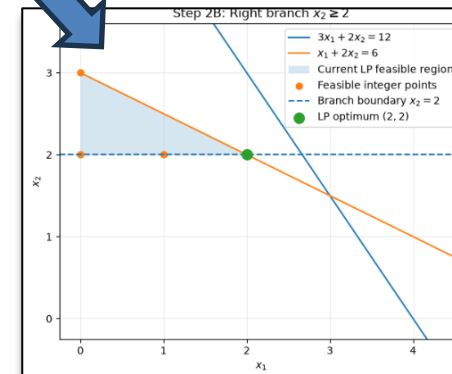
In summary:



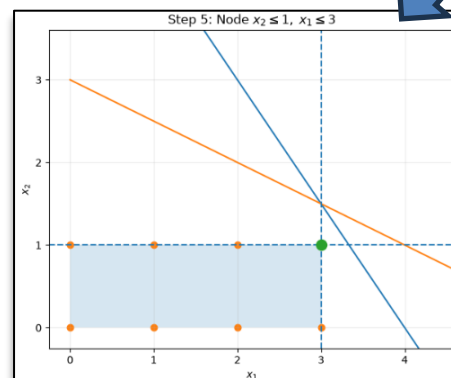
$$f(3, 1.5) = 27$$



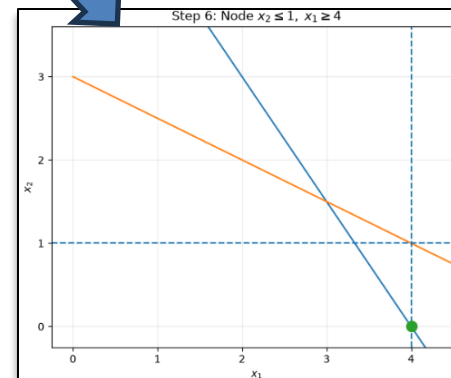
$$f(3.33, 1) = 20.67$$



$$f(2, 2) = 18$$



$$f(3, 1) = 19$$



$$f(4, 0) = 20$$

We stop because there are no unresolved (active) nodes left.

All solutions are already the best in each region.

Outline

1. Motivational Example
2. Branch-and-Bound
3. Examples in Energy Engineering
 - a. **Simple Unit Commitment**
 - b. **Unit Commitment with Renewables and Storage**
 - c. **Waste-to-Energy Facility Location-Allocation**

Unit Commitment: Simple

A very simple multi-period unit commitment problem is shown below:

$$\begin{aligned} \min_{P_{g,t}, u_{g,t}, y_{g,t}} \quad & \sum_t \sum_g c_g^{var} P_{g,t} + c_g^{NL} u_{g,t} + c_g^{SU} y_{g,t} \\ \text{s. t.} \quad & \sum_g P_{g,t} = D_t \\ & P_g^{min} u_{g,t} \leq P_{g,t} \leq P_g^{max} u_{g,t} \\ & u_{g,t} - u_{g,t-1} = y_{g,t} - z_{g,t} \\ & y_{g,t} + z_{g,t} \leq 1 \\ & P_{g,t} - P_{g,t-1} \leq R_g^{up} + P_g^{max} y_{g,t} \\ & P_{g,t-1} - P_{g,t} \leq R_g^{down} + P_g^{max} z_{g,t} \end{aligned}$$

* Assume a total period of 15 hrs and 1 period = 1 hr.

Decision Variables:

$P_{g,t}$ = power output of generator g at time t

$u_{g,t} \in \{0,1\}$ = indicates if gen g is online at time t

$y_{g,t} \in \{0,1\}$ = indicates if gen g undergoes startup at time t

$z_{g,t} \in \{0,1\}$ = indicates if gen g undergoes shutdown at time t

Other parameters:

c_g^{var} = variable generation cost of generator g

c_g^{NL} = no-load cost of generator g

c_g^{SU} = startup cost of generator g

P_g^{min}, P_g^{max} = min and max power output for generator g

D_t = demand at time t

Unit Commitment with Renewables and Storage

A 24-hour unit commitment problem is shown below:

$$\begin{aligned} \min_{P_{g,t}, u_{g,t}, y_{g,t}} \quad & \sum_t \left[\sum_g (c_g^{var} P_{g,t} + c_g^{SU} y_{g,t}) + c_B (P_{ch,t} + P_{dis,t}) \right] \\ \text{s.t.} \quad & P_{s,t} + P_{w,t} + P_{dis,t} + \sum_g P_{g,t} = D_t + P_{ch,t} \\ & 0 \leq P_{s,t} \leq S_t^{avail} \\ & 0 \leq P_{w,t} \leq W_t^{avail} \\ & P_g^{min} u_{g,t} \leq P_{g,t} \leq P_g^{max} u_{g,t} \\ & E_t = E_{t-1} + \eta_{ch} P_{ch,t} - \frac{P_{dis,t}}{\eta_{dis}} \\ & u_{g,t} - u_{g,t-1} = v_{g,t} - w_{g,t} \\ & P_{ch,t} \leq P_{ch}^{max} z_t \\ & v_{g,t} + w_{g,t} \leq 1 \\ & P_{dis,t} \leq P_{dis}^{max} (1 - z_t) \\ & P_{g,t} - P_{g,t-1} \leq R_g^{up} + P_g^{max} v_{g,t} \\ & 0 \leq E_t \leq E^{max} \\ & P_{g,t-1} - P_{g,t} \leq R_g^{down} + P_g^{max} w_{g,t} \\ & E_{24} = E^{init} \end{aligned}$$

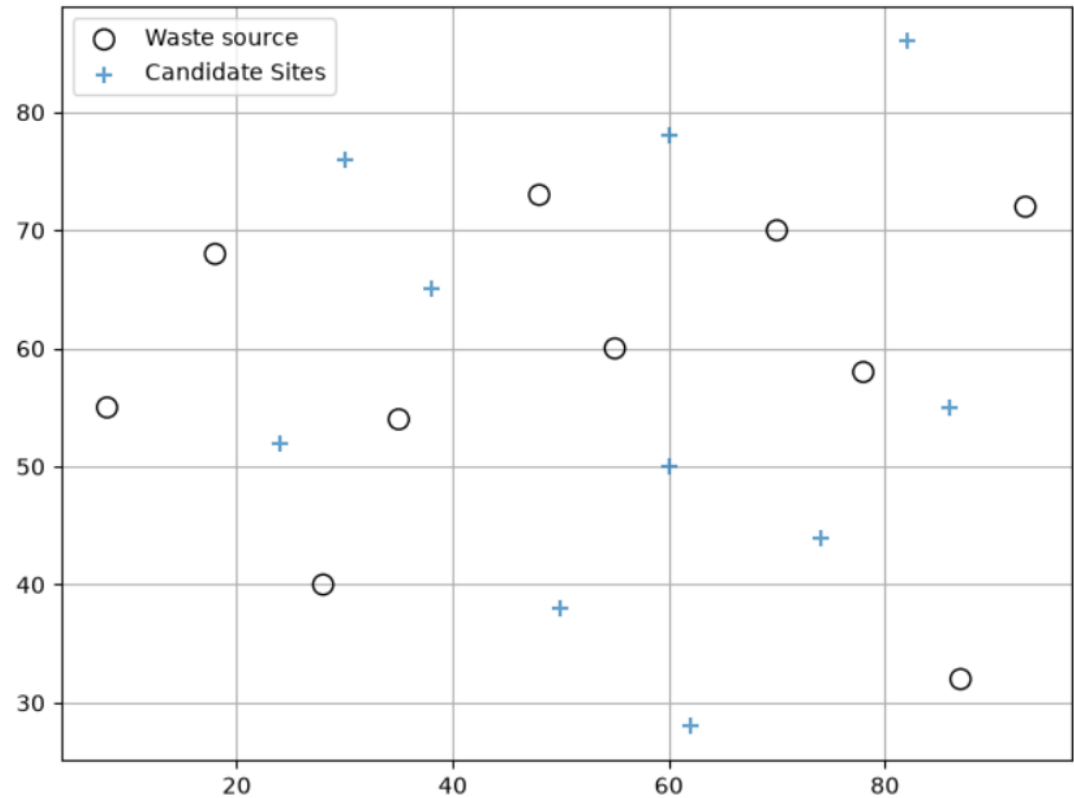
* Assume a total period of 24 hrs and 1 period = 1 hr.

Waste-to-Energy Facility Location-Allocation Planning

Our goal is to find the best location, best technology, and best transport of wastes in the following map:

$$\min_{z_{s,k}, a_{i,s}, Q_{s,k}} \quad CRF \sum_s \sum_k K_k z_{s,k} + \sum_s \sum_k c_k^{op} Q_{s,k} + c^{tr} \sum_i \sum_s W_i d_{i,s} a_{i,s} - p^E \sum_s \sum_k e_k Q_{s,k}$$

k	Technology
1	Pyrolysis
2	Incineration
3	Anaerobic Digestion



Decision Variables:

$z_{s,k} \in \{0,1\}$ = indicates if tech k is built at site s

$a_{i,s} \in \{0,1\}$ = indicates if waste source i is assigned to site s

$Q_{s,k}$ = annual waste processed at site s using tech k [ton/yr]

Fixed parameters:

CRF = capital recovery factor given discount rate r and 10-yr project life

K_k = upfront installation cost for tech k [\$]

c_k^{op} = operating cost for tech k [\$/ton]

c^{tr} = waste transportation cost [\$/ton-km]

e_k = net saleable energy yield [MWh/ton]

p^E = energy selling price [\$/MWh]

$d_{i,s}$ = straight-line distance from source i to site s

Outline

1. Motivational Example
2. Branch-and-Bound
3. Examples in Energy Engineering
 - a. Simple Unit Commitment
 - b. Unit Commitment with Renewables and Storage
 - c. Waste-to-Energy Facility Location-Allocation